Louis ALMAIRAC

Master MVA student at ENS Paris-Saclay and engineering student at École des Ponts, working on machine learning for time series and biosignals.

Seeking a 6-month research internship from April 2027.

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EDUCATION

École Normale Supérieure Paris-Saclay • *Paris, France* *2026 – 2027*
Master MVA (Mathematics, Vision, Learning)
Relevant courses: Time Series, Object Recognition, Foundations of LLMs, Computational Statistics, Convex Optimisation

École Nationale des Ponts et Chaussées • *Paris, France* *2022 – 2027*
MSc in Engineering: Mathematics and Computer Science
Relevant courses: Machine Learning, Stochastic Processes, Deep Learning, Statistics, Computer Vision, Programming

Lycée Pierre de Fermat, Classes Préparatoires (PCSI, PSI*) • *Toulouse, France* *2020 – 2022*

PUBLICATIONS

L. Almairac, M. Boukadoum. *EEG Decoding of In-Game Attentional Engagement.* IEEE ISCAS 2027, submitted.
L. Almairac, A. André, M. Pascal, A. Guillet, F. Argoul, P. Argoul. *Temporal variability of spectral Granger causality from neural networks (EEGs) during sleep: a time-frequency approach.* Poster, International Summer Institute on Network Physiology (ISINP), Lake Como, 2025.

EXPERIENCE

Research Intern, Université du Québec à Montréal • *Montréal, Canada* *March 2026 – July 2026*

- Decoded in-game attentional engagement from the EEG of 26 players (Ubisoft La Forge data) with Prof. Mounir Boukadoum; first-author paper submitted to IEEE ISCAS 2027.
- Designed a pipeline combining multivariate empirical mode decomposition, time-frequency analysis and Riemannian geometry (macro-F1 = 0.76, leave-one-subject-out), on par with an EEG-Conformer; code open-sourced on GitHub.
- Ran control analyses (ICA, eye-tracking-verified labels, permutation tests, self-supervised probes) showing that the decodable signal is a play-vs-rest arousal shift.

Machine Learning Intern, Caisse des Dépôts • *Paris, France* *June 2025 – Dec. 2025*

- In a systematic equity team (EUR 2bn AUM), implemented a LambdaMART learning-to-rank model selecting countries within an emerging-market universe, replacing a neural ListMLE baseline and optimising NDCG directly.
- Engineered the financial features (momentum, volatility, FX-equity correlation, drawdowns) and a purged, embargoed temporal cross-validation ruling out look-ahead bias.

Research Project, École des Ponts with CNRS and Max Planck • *Paris, France* *Nov. 2024 – June 2025*

- Developed a time-frequency Granger causality framework combining wavelet decomposition and VAR modelling to track directed interactions between EEG channels across sleep stages (PhysioNet 2018, 1,985 subjects).

Data Scientist Intern, Sopra Steria • *Paris, France* *Jan. 2024 – June 2024*

- Built the analysis tools identifying the causes of metro delays, for an RATP line-automation project.

PROJECTS & COMPETITIONS

Martha, AI Stylist App (personal project) *2025 – 2026*
Native iOS app (22k lines, SwiftUI, Supabase) that digitises a wardrobe and suggests outfits with vision-language models.

Medical Data Challenge, QRT *2025*
Survival prediction for leukaemia patients, with Institut Gustave Roussy; IPCW C-index of 0.755 (public leaderboard).

Hackathon, Mistral AI x Alan *2024*
Ranked 4th out of 100+ participants with an LLM-powered food-waste prevention app built in 24h.

Mathematics Competition, TFJM² *2020 and 2023*
2nd place nationally in 2020, in a team of 8; then on the organising team of the 2023 edition (community manager).

SKILLS & INTERESTS

Languages	French (native), English (fluent), Spanish (fluent)
Programming	Python (PyTorch, NumPy, pandas, scikit-learn, LightGBM, MNE, pyRiemann), Swift, Git
Hobbies	Competitive Rugby player, Member of English Debating Club